Hyunjae Kim

present@kaist.ac.kr
Hyunjae Kim
present

Education

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Republic of Korea
Ph.D. in Culture Technology (Graduate School of Culture Technology) — Mar. 2022 – Expected Feb. 2027

Chungnam National University — Daejeon, Republic of Korea
M.S. in Physics — Mar. 2018 – Feb. 2020

- Thesis: "Study of speech signal processing of cochlea with bio-inspired system based on artificial neural network"

Chungnam National University — Daejeon, Republic of Korea
B.S. in Physics — Mar. 2011 – Feb. 2018

Research & Professional Experience

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Republic of Korea
Ph.D. Student, Music and Brain Lab (Advisor: Kyung Myun Lee) — Mar. 2022 –Present

- Investigating how spectrotemporal modulation constraints can give rise to human-like cortical music selectivity through deep learning-based computational modeling.
- Investigating representational alignment of deep audio models with human brain responses.
- Conducting human EEG and behavioral experiments using spatial audio and naturalistic music to investigate neural correlates of immersive experience and music preference.

Deep Hearing Corp. — Daejeon, Republic of Korea
Chief Operating Officer / AI Developer — Aug. 2020 – Nov. 2021

- Managed and contributed to the development of low latency deep learning models for real-time speech enhancement.

Chungnam National University — Daejeon, Republic of Korea
Researcher, Biomimetic Machine Learning Lab — Mar. 2020 –Jul. 2020
M.S. Student, Biomimetic Machine Learning Lab (Advisor: Kang-Hun Ahn) — Mar. 2018 –Feb. 2020

- Built a physical cochlea-mimicking system with a deep learning back end for signal processing.
- Investigated deep learning approaches for speech enhancement and speech recognition.

Publications

1. **H Kim**, S-B Paik, KM Lee. *Deep Auditory Models with Temporal Modulation Constraints Reveal the Emergence of Human-like Music Selectivity.* Manuscript in preparation.
2. **H Kim**, KM Lee. *Brain Alignment of Music Foundation Models: Do Better Representations Mirror the Auditory Cortex?* Under review.
3. **H Kim**, R Sato, S Kim, KM Lee. *Neural Correlates of Immersive Spatial Audio: Enhanced Neural Synchrony and Reduced Late Cortical Responses During Naturalistic Listening.* Under review.
4. **H Kim**, J Nam, KM Lee. (2026). Transferable Audio Lottery Tickets: Gradient Accumulation for Extreme Sparsity. *In Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP).*

5. T Song, **H Kim**, KM Lee. (2026). Musical Emotion Under Cochlear Implant Listening: Human Perception and Foundation Model Representations of Human-Composed and AI-Generated Music. *In Proceedings of the International Society for Music Information Retrieval Conference (ISMIR).*

6. M Bae, **H Kim**, KM Lee. (2026). Gender Bias and Sonic Gender Performance in AI-Generated Music: Representation, Expression, and Perception. *In Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES).*

7. S Cho, **H Kim**, KM Lee. (2026). Predicting Music Preference Using Deep Learning on Electroencephalogram and Audio Features. *The Journal of the Acoustical Society of Korea.*

8. EJ Oh, **H Kim**, KM Lee. (2024). Which audio features can predict the dynamic musical emotions of both composers and listeners? *In Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR).*

9. S Park, **H Kim**, KM Lee. (2024). Enhancing Predictive Models of Music Familiarity with EEG: Insights from Fans and Non-fans of K-pop Group NCT127. *In Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR).*

10. WS Lee, **H Kim**, AN Cleland, KH Ahn. (2021). Fast frequency discrimination and phoneme recognition using a biomimetic membrane coupled to a neural network. *Bioinspiration & Biomimetics, 16(2), 026012.*

11. **H Kim**, WS Lee, J Yoo, M Park, KH Ahn. (2019). Origin of the higher difficulty in the recognition of vowels compared to handwritten digits in deep neural networks. *Journal of the Korean Physical Society, 74, 12–18.*

Conference Abstracts

2026

- **H Kim**, KM Lee. Do AI Models Encode Music Like Humans? Representational Alignment in Music AI. *Asia-Pacific Society for the Cognitive Sciences of Music (APSCOM).*

2025

- **H Kim**, J Nam, KM Lee. Transferable Audio Lottery Tickets: Gradient Accumulation for Extreme Sparsity. *Korean Society for Music Informatics.*
- **H Kim**, R Sato, S Kim, KM Lee. Neural Correlates of Immersive Audio Experience: An EEG Study with TRF and ISC in Naturalistic Listening. *Acoustical Society of America (ASA).*
- **H Kim**, R Sato, S Kim, KM Lee. Comparison of Neural Responses to Immersive and Stereo Audio: An EEG-Based Analysis Under Naturalistic Listening Conditions. *Korean Society of Music Perception and Cognition (KSMPC).*
- M Lee, **H Kim**, KM Lee. How Do People Evaluate AI-Generated Music? A Quantitative Analysis of the Relationship Between Musical Features and Multidimensional Human Judgments. *Korean Society of Music Perception and Cognition (KSMPC).*
- M Bae, **H Kim**, KM Lee. Does AI Generate "Masculine" Music? An Analysis of Gender Bias in AI-Generated Music. *Korean Society of Music Perception and Cognition (KSMPC).*
- S Cho, **H Kim**, KM Lee. Predicting Music Preference Using Deep Learning on EEG and Audio Features. *Korean Society of Music Perception and Cognition (KSMPC).*

2024

- **H Kim**, EJ Choi, EJ Oh, J Nam, KM Lee. Evaluation of AI-Generated Music through Online Surveys. *67th Korean Society of Music Perception and Cognition (KSMPC).*

2023

- **H Kim**, EJ Oh, J Park, Y Chung, J Nam, KM Lee. A Comparison of Melody Interpolation Performed by Human and Artificial Intelligence Based on Human Similarity Judgments. *International Conference on Music Perception and Cognition (ICMPC).*
- **H Kim**, EJ Oh, Y Chung, J Nam, KM Lee. A Study of Melody Similarity of AI Composition with a Cognitive Behavioral Experiment. *66th Korean Society of Music Perception and Cognition (KSMPC).*

2017

- **H Kim**, Y Seong, M Kim, M Park, J Yoo, W Lee, KH Ahn. Differences of Deep Learning Models in Speech Recognition and Text Recognition. *Korean Physical Society Fall Meeting.*

Teaching

Korea Advanced Institute of Science and Technology (KAIST)

- Mentor: Undergraduate Research Participation (URP) program — Fall 2026
- TA: Understanding Music and Brain — Fall 2024
- Mentor: Undergraduate Research Participation (URP) program — Fall 2024
- TA: Cognitive Science of Music — Spring 2024
- Mentor: Undergraduate Research Participation (URP) program — Spring 2024
- TA: Understanding Music and Music History — Fall 2023
- TA: Cognitive Science of Music — Spring 2023
- TA: Understanding Music and Music History — Fall 2022
- TA: Social Network Analysis — Spring 2022

Chungnam National University

- TA: General Physics Experiments — Spring 2019
- TA: General Physics Experiments — Fall 2018
- TA: General Physics Experiments — Spring 2018

Scholarships and Awards

- First Place, CBrain Team Project Competition (Korean Society for Computational Neuroscience) — 2026
- Academic Excellence Scholarship (Chungnam National University) — 2019
- Academic Excellence Scholarship (Chungnam National University) — 2018
- Outstanding Presentation Award (Korean Physical Society) — 2017
- Academic Merit Scholarship (Chungnam National University) — 2013-2017
- Full Tuition Admission Scholarship (Chungnam National University) — 2011
  Awarded for entering with the highest academic distinction.

Patent

- K. H. Ahn, **H. Kim**, "Apparatus for adjusting output sound source and method thereof", KR 10-2020-0088942 (Application)
- K. H. Ahn, **H. Kim**, N. Boichenko, "Method and apparatus for gene expression level prediction", KR 10-2021-0021143 (Granted)